

IDS Pipeline - Intrusion Detection System with ML Enhancement

A high-performance Intrusion Detection System combining Suricata, Kafka, and Machine Learning for real-time threat detection and analysis.

🔗 Documentation

- 📖 [PIPELINE_ARCHITECTURE.md](#) - Detailed explanation of how both AF_PACKET and DPDK pipelines work
- 🚀 [NEXT_STEPS.md](#) - Roadmap and future development plans
- 📝 [CLEANUP_REPORT.md](#) - Recent codebase cleanup details

🚀 Quick Start

Choose Your Mode:

1. AF_PACKET Mode (Recommended for most users)

- Works with ANY network interface (including USB adapters)
- No special drivers required
- Easy setup

```
sudo ./run_afpacket_mode.sh
```

2. DPDK Mode (High-performance)

- Requires DPDK-compatible NIC
- Kernel bypass for maximum throughput
- More complex setup

```
sudo ./run_dpdk_mode.sh
```

📁 Architecture

```
Network Interface → Suricata IDS → Kafka → ML Consumer → Enhanced Alerts
                    ↓
                Signature Detection
                    ↓
                Flow Extraction
                    ↓
                ML Inference
```

Prerequisites

Required Software:

- **Suricata** (IDS engine)
- **Apache Kafka** (Message broker)
- **Python 3.8+** (ML processing)
- **tcpreplay** (Traffic replay - optional)

For AF_PACKET Mode:

- Any Linux network interface
- No special drivers needed

For DPDK Mode:

- DPDK-compatible NIC (Intel, Mellanox, etc.)
- Suricata compiled with DPDK support
- DPDK libraries installed
- Hugepages configured (2GB recommended)

Installation

1. Install Dependencies

```
# Update system
sudo apt update && sudo apt upgrade -y

# Install Suricata
sudo apt install suricata -y

# Install Kafka (if not installed)
wget https://downloads.apache.org/kafka/3.6.0/kafka_2.13-3.6.0.tgz
tar -xzf kafka_2.13-3.6.0.tgz
sudo mv kafka_2.13-3.6.0 /usr/local/kafka

# Install Python dependencies
pip install -r requirements.txt

# Install tcpreplay (optional, for traffic replay)
sudo apt install tcpreplay -y
```

2. Configure the Pipeline

Edit the configuration file:

```
nano dpdk_suricata_ml_pipeline/config/pipeline.conf
```

Key settings:

- **NETWORK_INTERFACE** - Your network interface name
- **ML_MODEL_PATH** - Path to your ML model
- **KAFKA_BOOTSTRAP_SERVERS** - Kafka server address

3. Choose Your Mode

For AF_PACKET Mode:

```
sudo ./run_afpacket_mode.sh start
```

For DPDK Mode:

```
sudo ./run_dpdk_mode.sh start
```

Usage Guide

AF_PACKET Mode

The AF_PACKET script provides an interactive menu:

```
sudo ./run_afpacket_mode.sh
```

Menu Options:

1. Start Complete Pipeline - Starts all components
2. Start Kafka Only
3. Start Suricata Only
4. Start ML Consumer Only
5. Start Kafka Bridge Only
6. Replay Traffic - Replay PCAP files
7. Check Status - View system status
8. View Logs - Monitor logs in real-time
9. Setup External Capture - Configure for external traffic
10. Stop All Services

Command-line Usage:

```
sudo ./run_afpacket_mode.sh start    # Start everything
sudo ./run_afpacket_mode.sh status  # Check status
sudo ./run_afpacket_mode.sh stop    # Stop all services
sudo ./run_afpacket_mode.sh logs    # View logs
```

DPDK Mode

The DPDK script includes additional DPDK-specific options:

```
sudo ./run_dpdk_mode.sh
```

Menu Options:

1. Start Complete Pipeline
2. Start Kafka Only
3. Start Suricata Only (DPDK)
4. Start ML Consumer Only
5. Start Kafka Bridge Only
6. Bind Interface to DPDK - Bind NIC to DPDK driver
7. Unbind Interface from DPDK - Restore NIC to kernel
8. Check Status
9. View Logs
10. Show DPDK Info - Display DPDK device status
11. Stop All Services

Command-line Usage:

```
sudo ./run_dpdk_mode.sh start    # Start everything
sudo ./run_dpdk_mode.sh bind    # Bind interface to DPDK
sudo ./run_dpdk_mode.sh unbind  # Unbind interface
sudo ./run_dpdk_mode.sh status  # Check status
sudo ./run_dpdk_mode.sh stop    # Stop all services
```

🔍 Components

1. Suricata IDS

- Monitors network traffic
- Applies signature-based detection rules
- Generates flow and alert data
- Outputs to EVE JSON format

2. Kafka Message Broker

- Receives Suricata events
- Provides reliable message queuing
- Enables distributed processing
- Topics: `suricata-alerts`, `ml-predictions`

3. Suricata-Kafka Bridge

- Reads Suricata EVE JSON logs
- Publishes events to Kafka
- Real-time streaming
- Auto-reconnection

4. ML Consumer

- Consumes events from Kafka
- Extracts CICIDS2017 features (65 features)
- Maps to model format (34 features)
- Performs ML inference
- Combines with Suricata alerts
- Publishes enhanced alerts

Features

ML-Enhanced Detection

- **Flow-based analysis** - Every network flow analyzed by ML
- **65 CICIDS2017 features** - Industry-standard feature set
- **Multiple models** - Random Forest, LightGBM support
- **Real-time inference** - Low-latency predictions
- **Confidence scoring** - Probabilistic threat assessment

Performance

- **AF_PACKET Mode:** ~1-10 Gbps depending on hardware
- **DPDK Mode:** 10+ Gbps with kernel bypass
- **Scalable:** Kafka enables horizontal scaling
- **Efficient:** Batch processing for ML inference

Project Structure

```
IDS/
├── run_afpacket_mode.sh      # AF_PACKET mode master script
├── run_dpdk_mode.sh         # DPDK mode master script
├── cleanup_codebase.sh      # Cleanup redundant files
├── requirements.txt          # Python dependencies
├── README.md                # This file
├──
└── config/
```

```

├── ids_config.yaml          # ML feature configuration
├── dpdk_suricata_ml_pipeline/
│   ├── config/
│   │   └── pipeline.conf    # Pipeline configuration
│   ├── scripts/            # Individual component scripts
│   │   ├── 00_setup_external_capture.sh
│   │   ├── 01_bind_interface.sh
│   │   ├── 02_setup_kafka.sh
│   │   ├── 03_start_suricata.sh      (DPDK)
│   │   ├── 03_start_suricata_afpacket.sh (AF_PACKET)
│   │   ├── 04_start_ml_consumer.sh
│   │   ├── 05_replay_traffic.sh
│   │   ├── 06_start_kafka_bridge.sh
│   │   └── stop_all.sh
│   ├── src/                # Python source code
│   │   ├── ml_kafka_consumer.py
│   │   ├── feature_extractor.py
│   │   ├── feature_mapper.py
│   │   ├── model_loader.py
│   │   └── alert_processor.py
│   ├── logs/               # Log files
│   ├── models/             # ML models
│   └── pcap_samples/       # Sample PCAP files
├── ML Models/              # Pre-trained models
│   ├── random_forest_model_2017.joblib
│   └── lgb_model_2018.joblib
├── notebooks/              # Jupyter notebooks
│   ├── CICIDS2017.ipynb
│   └── CICIDS2018.ipynb
└── tests/                  # Test scripts
    ├── quick_attack_demo.py
    └── quick_dpdk_test.py

```

Configuration

Pipeline Configuration ([dpdk_suricata_ml_pipeline/config/pipeline.conf](#))

```

# Network Interface
NETWORK_INTERFACE="eth0"          # Your interface name
INTERFACE_PCI_ADDRESS=""         # Auto-detect (DPDK only)
DPDK_DRIVER="vfio-pci"           # DPDK driver

# DPDK Settings

```

```
DPDK_HUGEPAGES="2048"           # 2GB hugepages
DPDK_CORES="0,1"                 # CPU cores

# Suricata
SURICATA_CONFIG="/etc/suricata/suricata-dpdk.yaml"
SURICATA_CORES="2"               # Worker threads
SURICATA_HOME_NET="192.168.0.0/16" # Your network

# Kafka
KAFKA_BOOTSTRAP_SERVERS="localhost:9092"
KAFKA_TOPIC_ALERTS="suricata-alerts"
KAFKA_TOPIC_ML_PREDICTIONS="ml-predictions"

# ML Model
ML_MODEL_PATH="/path/to/model.joblib"
ML_CONFIDENCE_THRESHOLD="0.7"    # Alert threshold
```

Logs

Log Locations:

- **Suricata:** `/var/log/suricata/suricata.log`
- **Suricata EVE JSON:** `/var/log/suricata/eve.json`
- **ML Consumer:** `dpdk_suricata_ml_pipeline/logs/ml/ml_consumer.log`
- **Kafka Bridge:** `dpdk_suricata_ml_pipeline/logs/kafka_bridge.log`

View Logs:

```
# Via menu
sudo ./run_afpacket_mode.sh
# Select option 8

# Direct access
tail -f /var/log/suricata/eve.json
tail -f dpdk_suricata_ml_pipeline/logs/ml/ml_consumer.log
```

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Testing

Replay Sample Traffic:

```
sudo ./run_afpacket_mode.sh
# Select option 6 (Replay Traffic)

# Or directly:
sudo tcpreplay -i eth0 -M 10 pcap_samples/sample.pcap
```

Quick Attack Demo:

```
cd tests
python3 quick_attack_demo.py
```



Troubleshooting

Suricata Won't Start

```
# Check Suricata config
sudo suricata -T -c /etc/suricata/suricata.yaml

# Check interface is up
ip link show eth0
```

Kafka Connection Issues

```
# Check Kafka is running
pgrep -f kafka

# Test Kafka connectivity
kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic
suricata-alerts
```

DPDK Binding Issues

```
# Check DPDK devices
sudo dpdk-devbind.py --status

# Unbind and retry
sudo ./run_dpdk_mode.sh unbind
sudo ./run_dpdk_mode.sh bind
```

ML Consumer Errors

```
# Check Python dependencies
pip install -r requirements.txt

# Verify model path
ls -la /path/to/model.joblib
```



```
# Check logs
tail -f dpdk_suricata_ml_pipeline/logs/ml/ml_consumer.log
```

Cleanup

To remove redundant files and documentation:

```
sudo ./cleanup_codebase.sh
```

This will:

- Remove all duplicate PDF files
- Remove redundant documentation (30+ files)
- Remove legacy code directory
- Backup everything before deletion

Additional Documentation

- **QUICKSTART.md** - Quick start guide
- **SETUP_GUIDE.md** - Detailed setup instructions
- **PRODUCTION_DPDK_GUIDE.md** - DPDK production deployment
- **REALTIME_PIPELINE_GUIDE.md** - Real-time processing guide
- **EXTERNAL_TRAFFIC_GUIDE.md** - External traffic capture setup
- **USB_ADAPTER_GUIDE.md** - USB network adapter usage
- **REMOTE_DEVICE_SETUP.md** - Remote device configuration

Contributing

Contributions are welcome! Please follow these steps:

1. Fork the repository
2. Create a feature branch
3. Make your changes
4. Test thoroughly
5. Submit a pull request

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License

This project is licensed under the MIT License.

Authors

- **Sujay** - Initial work

Acknowledgments

- Suricata IDS team
- Apache Kafka project
- CICIDS2017 dataset creators
- DPDK community

Support

For issues and questions:

- Check the troubleshooting section
- Review log files
- Open an issue on GitHub

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